GARRF VIRTUAL STRUCTURAL HEALTH MONITORING LABORATORY

STUDENT HELP & EXPERIMENT GUIDE

Piezoelectric (PZT) Based EMI & Guided Wave Structural Health Monitoring

Developed by

Dr. Annamdas Venu Gopal Madhav &

Annamdas Shantanu Vasudev Krishna

Gopalkrishna Advanced Rural Research Foundation (GARRF)

WELCOME TO THE GARRF VIRTUAL LABORATORY

This virtual laboratory has been designed to let you:

Explore → Experiment → Observe → Compare → Analyze → Understand

You are not expected to understand everything before starting.

Try the controls. Change the parameters. Watch the waves. Observe the waveform. Study the FFT. Introduce simulated damage. Compare the results.

The purpose is to help you develop an intuitive understanding of **Piezoelectric (PZT) based Structural Health Monitoring (SHM)**.

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1. WHY THIS VIRTUAL LABORATORY?

This type of Structural Health Monitoring laboratory is commonly available in leading international universities and advanced research laboratories.

A few IITs and premier research institutes in India are actively working in the area of **Piezoelectric (PZT) based Structural Health Monitoring**.

The **Gopalkrishna Advanced Rural Research Foundation (GARRF)** is bringing this type of advanced learning to rural students **free of cost**, so that geography, financial limitations and lack of access to expensive laboratory equipment do not become barriers to engineering education.

The objective is to give students an opportunity to explore concepts that they may otherwise encounter only in advanced university laboratories or research institutions.

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2. IMPORTANT NOTE ABOUT THE VIRTUAL BEAM AND PLATE

The **Beam** and **Plate** in this laboratory are **generic educational virtual structures**.

They do not currently represent a particular real-world beam or plate with a complete set of physical properties.

For example, the present generic model does not define a specific:

- Length
- Width
- Thickness
- Material grade
- Young's modulus
- Poisson's ratio
- Density
- Damping
- Boundary condition
- PZT size
- PZT material specification
- Adhesive properties
- Real crack geometry

Therefore:

The Beam and Plate are provided for conceptual understanding and experimentation, not as calibrated models of a particular real engineering structure.

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3. WHY ARE THE PROPERTIES IMPORTANT?

In a real engineering experiment, the response depends on the actual physical characteristics of the structure and sensor.

Changing the:

- Geometry
- Material
- Thickness

- Boundary conditions
- PZT properties
- PZT location
- Adhesive layer
- Excitation
- Environmental conditions

can change the measured response.

Consequently:

Actual results from a real beam or plate will be different from the results generated by this generic virtual laboratory.

The numerical values, waveforms, FFT peaks and damage indicators from this simulation should therefore **not be treated as universal engineering values**.

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4. WHAT IS THE MAIN THING YOU SHOULD LEARN?

The most important concept is the complete SHM chain:

PZT Excitation

↓

Structural Waves

↓

Wave Propagation

↓

Interaction with Damage

↓

Reflection / Scattering / Attenuation

↓

PZT Sensing

↓

Signal Acquisition

↓

Time-Domain Analysis

↓

FFT / Frequency-Domain Analysis

↓

Damage Identification

This is the fundamental idea behind the experiment.

PART 1 — SET UP YOUR EXPERIMENT

Step 1 — Select the Structure

Choose:

Beam

or

Plate

Remember that these are generic educational structures.

Step 2 — Position the PZT

Drag the PZT patch to the desired location.

Observe its position relative to the structure.

For a controlled Healthy-versus-Damaged experiment, keep the PZT position unchanged.

Later, move the PZT and conduct another experiment to see how sensor position affects the response.

Step 3 — Select Excitation Frequency

Choose an excitation frequency.

Different frequencies can interact differently with structural modes and damage conditions.

You do not need to understand this completely before experimenting.

Try different frequencies and observe what happens to the FFT.

Step 4 — Select Excitation Type

You can select:

SINE

Single-frequency excitation.

Useful for studying frequency-specific response.

SWEEP

Changing frequency excitation.

Useful for exploring a broader frequency response.

PULSE

Short excitation.

Useful for understanding wave propagation, reflections and time-domain behavior.

PART 2 — SELECT MONITORING DURATION

You can choose:

1 MINUTE

2 MINUTES

or

5 MINUTES

For your first experiment, choose:

1 MINUTE

Later, try 2 minutes and 5 minutes.

IMPORTANT — THE TIMER IS DELIBERATELY EXPLORATORY

The monitoring-duration options are provided to **guide your experiment**, but the current learning version deliberately allows the experiment to continue rather than forcing an automatic stop exactly at the selected time.

This is intentional.

The idea is to encourage you to:

observe → discover → explore

rather than simply wait for the software to terminate the experiment.

Therefore:

If you select 1 minute and the experiment continues, don't worry.

Monitor for approximately 1 minute and press:

STOP

manually.

The same applies to 2 minutes and 5 minutes.

Example

1 minute selected

→ Observe for approximately 1 minute

→ Press STOP if it is still running

Or

2 minutes selected

→ Observe for approximately 2 minutes

→ Press STOP if it is still running

Or

5 minutes selected

→ Observe for approximately 5 minutes

→ Press STOP if it is still running

There is nothing wrong if you need to press STOP manually.

PART 3 — START WITH HEALTHY

Always begin with:

HEALTHY

This establishes your reference condition.

It is your:

Healthy Baseline or

Stage 0

Healthy Experiment

1. Select **Healthy**.
2. Confirm that Severity is locked.
3. Select the monitoring duration.
4. Press **START**.
5. Watch the structure.
6. Watch the simulated waves.
7. Watch the live waveform.
8. Watch the FFT.
9. Observe for your selected duration.
10. Press **STOP** manually if necessary.

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5. WHY IS SEVERITY LOCKED WHEN HEALTHY IS SELECTED?

This is intentional.

When the structure is:

HEALTHY

there is no simulated damage.

Therefore there is no damage severity to specify.

The interface should show:

Severity: N/A — Healthy

and keep the control locked.

When you select a damage condition, Severity becomes active.

PART 4 — INTRODUCE SIMULATED DAMAGE

After exploring Healthy, select one of the available damage conditions.

Possible conditions include:

- **Crack**

- **Debonding**
- **Loosened Connection**
- **Local Stiffness Reduction**

CRACK

Represents a localized discontinuity in the virtual structure.

It can change simulated wave propagation, reflection and scattering.

DEBONDING

Represents a simulated weakening or loss of the PZT/structure interface response.

This is important because sensor bonding can also influence PZT measurements in real experiments.

LOOSE CONNECTION

Represents a simulated change in structural connection or boundary stiffness.

LOCAL STIFFNESS REDUCTION

Represents a simulated localized reduction in structural stiffness.

PART 5 — UNDERSTANDING SIMULATION SEVERITY

When you select a damage condition, Severity becomes active.

For example:

Crack — Severity 30%

or

Crack — Severity 60%

Use Severity to explore how changing the simulated damage parameter affects the response.

IMPORTANT

Severity is a **simulation parameter**.

It does not automatically mean:

- 30% crack length
- 30% structural failure
- 30% material failure
- 30% loss of strength

unless a validated physical model specifically establishes that relationship.

Therefore, think of it as:

Simulation Severity

PART 6 — RUN THE DAMAGED EXPERIMENT

For your first comparison, use:

Healthy → Crack

Keep the following unchanged:

- Structure
- PZT position
- Frequency
- Excitation type
- Monitoring duration

Change only:

Healthy → Crack

Then:

1. Select **Crack**.
2. Set the Simulation Severity.
3. Press **START**.
4. Watch the structure.
5. Watch the waves.
6. Watch the waveform.
7. Watch the FFT.

8. Observe for your selected duration.
9. Press **STOP** manually if necessary.

You now have:

Healthy

versus

Simulated Crack

PART 7 — UNDERSTAND THE WAVE VISUALIZATION

The moving wave represents the simulated mechanical disturbance travelling through the virtual structure.

When the wave encounters a simulated structural change, you may observe:

- Reflection
- Scattering
- Attenuation
- Local vibration changes
- Changes in the wave pattern

The objective is to develop an intuitive understanding of how a structural change can influence wave propagation.

PART 8 — UNDERSTAND THE LIVE WAVEFORM

The waveform represents the simulated PZT response as a function of time.

Compare:

Healthy waveform

with

Damaged waveform

Look for changes in:

- Amplitude
- Shape
- Peaks

- Periodicity
- Energy
- Arrival time
- Reflections

Remember:

A difference in a signal does not automatically prove physical damage.

PART 9 — UNDERSTAND THE FFT

FFT converts the time-domain signal into a frequency-domain representation.

Look for:

- Peak frequency
- Peak amplitude
- New peaks
- Reduced peaks
- Peak shifts
- Changes in spectral shape

Damage may change the frequency-domain response because the structural mechanical response has changed.

PART 10 — WHY DOES THE PZT RESPONSE CHANGE?

The basic physical process is:

PZT

↓

Excites the structure

↓

Mechanical wave travels

↓

Wave encounters a structural condition

↓

Wave is reflected / scattered / attenuated

↓

Changed response reaches the PZT

↓

PZT produces a changed electrical response

Therefore:

A change in the measured electrical response can contain information about a change in structural condition.

This is one of the fundamental concepts behind PZT-based SHM.

PART 11 — HEALTHY VS DAMAGED COMPARISON

Ask yourself:

Wave pattern

Did the simulated propagation change?

Amplitude

Did the response become stronger or weaker?

Waveform

Did the time-domain shape change?

FFT

Did a frequency peak move or change amplitude?

RMS

Did the overall signal magnitude change?

Correlation

How similar are the two signals?

Damage Indicator

Did the laboratory-defined indicator show a larger difference?

PART 12 — CHANGE ONE VARIABLE AT A TIME

This is very important.

If you want to study the effect of a crack:

Keep constant:

Beam

PZT position

Frequency

Sine

1 minute

Change only:

Healthy → Crack

This makes the comparison easier to understand.

Once you understand the basic experiment, deliberately change other parameters and explore their effects.

PART 13 — EXPLORE DIFFERENT EXPERIMENTS

Try:

Experiment 1

Healthy → Crack

Experiment 2

Crack Severity 20% → Crack Severity 60%

Experiment 3

Sine → Sweep

Experiment 4

Pulse excitation

Experiment 5

Change frequency

Experiment 6

Move the PZT

Experiment 7

Healthy → Debonding

Experiment 8

Healthy → Loosened Connection

Experiment 9

Healthy → Local Stiffness Reduction

The purpose is to discover how different conditions influence the simulated response.

PART 14 — EXPLORE DIFFERENT MONITORING TIMES

Try:

1 minute

then:

2 minutes

then:

5 minutes

Use the additional time to observe:

- Signal stability
- Repeating patterns
- Wave behavior
- Waveform behavior
- FFT behavior
- Changes during continued excitation

Again:

If the experiment does not stop automatically, don't worry.

Press **STOP** manually after the intended monitoring period.

PART 15 — IF YOU PRESS STOP

Manual STOP is allowed.

If the experiment is still running after your intended monitoring period:

Press STOP.

You have not done anything wrong.

For a controlled comparison, try to use approximately the same observation duration for Healthy and Damaged conditions.

For example:

Healthy → 1 minute → STOP

↓

Crack → 1 minute → STOP

↓

Compare

PART 16 — SAVING YOUR EXPERIMENT FOR ARJUN

This is an important part of the learning process.

After completing your experiment, use the laboratory's **Download / Save Experiment Data** function.

Save the experiment file to your own computer.

Do not assume that the file is automatically transferred to ARJUN.

The virtual laboratory and ARJUN are separate environments.

The laboratory is hosted separately, while ARJUN operates as your AI analysis assistant.

Therefore, the student should learn the complete data workflow:

Conduct Experiment

↓

Observe

↓

Complete / Stop

↓

Download Experiment Data

↓

Save File on Your Computer

↓

Open ARJUN

↓

Upload the Saved File

↓

Ask ARJUN to Analyze

This manual transfer is **deliberate**.

It helps students understand an important real-world engineering workflow:

Measurement → Data File → Data Transfer → Analysis → Interpretation

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PART 17 — WHY DO WE MAKE YOU SAVE THE FILE FIRST?

The purpose is educational.

In a real laboratory, data is normally acquired by an instrument and stored as a file or dataset.

The engineer then takes that data into another analysis environment.

The GARRF Virtual Laboratory is designed to help students understand this process.

Therefore, do not think:

"Why doesn't the virtual laboratory automatically send everything to ARJUN?"

Instead think:

"I generated experimental data, saved it, and transferred it to an analysis system."

That is a valuable engineering habit.

PART 18 — USING ARJUN AFTER THE SIMULATION

After saving your experiment file:

1. Open **ARJUN**.
2. Upload the saved experiment file.

3. Tell ARJUN that it came from the **GARRF Virtual Structural Health Monitoring Laboratory**.
4. Ask ARJUN to analyze the experiment.
5. Ask questions about the waveform, FFT and damage indicator.
6. Ask ARJUN to explain the physical meaning in simple engineering language.

For example:

"This file was generated from the GARRF Virtual SHM Laboratory. I first ran Healthy and then Crack. Please analyze the difference."

ARJUN can then interpret the information provided in the file.

PART 19 — ARJUN CAN ALSO WORK STANDALONE

ARJUN is not limited to the virtual laboratory.

You can also use ARJUN independently for:

Real experimental data

from:

- University laboratories
- Research laboratories
- Industry
- Field experiments
- Oscilloscopes
- Impedance analyzers
- Network analyzers
- PZT experiments
- EMI experiments

You can provide:

- CSV
- Excel
- PDF
- Images

- Screenshots
- Waveforms
- Frequency-domain data
- Time-domain data

ARJUN should first understand whether the data is:

GARRF Virtual Laboratory Simulation

or

Real Experimental Data

before interpreting the result.

PART 20 — IMPORTANT FOR REAL EXPERIMENTS

The generic virtual model should not be confused with a real physical structure.

If you perform a real experiment on a beam with known:

- Dimensions
- Material
- Thickness
- Boundary conditions
- PZT properties
- PZT location
- Adhesive properties
- Damage location
- Damage size

the resulting signal will generally be different.

ARJUN should therefore distinguish between:

Generic Virtual Laboratory Data

and

Physically Characterized Experimental Data

This distinction is essential for responsible engineering analysis.

PART 21 — WHAT ARJUN SHOULD NOT CLAIM

A simulated result should not be presented as proof of a real structural defect.

For example:

Virtual simulation: Crack condition detected

does not mean:

A real beam has been proven to contain a crack.

Real structural assessment requires appropriate instrumentation, validation, engineering judgment and physical testing.

PART 22 — YOUR FIRST COMPLETE LEARNING PATH

Follow this sequence:

1. Select Beam

↓

2. Position PZT

↓

3. Select Sine

↓

4. Select a frequency

↓

5. Select Healthy

↓

6. Severity remains locked

↓

7. Select 1 minute

↓

8. START

↓

9. Watch waves + waveform + FFT

↓

10. STOP after approximately 1 minute if necessary

↓

11. Save the experiment data

↓

12. Select Crack

↓

13. Select Simulation Severity

↓

14. START

↓

15. Watch waves + waveform + FFT

↓

16. STOP after approximately 1 minute if necessary

↓

17. Save the experiment data

↓

18. Compare Healthy vs Crack

↓

19. Open ARJUN

↓

20. Upload the saved file

↓

21. Tell ARJUN that it is GARRF Virtual Laboratory data

↓

22. Ask ARJUN to analyze it

PART 23 — TROUBLESHOOTING

Nothing moves

Refresh the latest HTML laboratory and try START again.

Severity is disabled

This is intentional.

You have selected:

Healthy

Select a damage condition to activate Severity.

The experiment does not stop at 1 minute

Don't worry.

This is deliberately allowed in the current learning version.

Observe for approximately 1 minute and press:

STOP

manually.

The same applies to:

2 minutes

and

5 minutes.

I pressed STOP manually

That is okay.

You have not made a mistake.

Continue your experiment or repeat the stage if you want a controlled run.

I cannot see the simulated crack

Select:

Crack

and press START.

Make sure you are not still in Healthy mode.

FFT does not change much

Check that you:

- Ran the Healthy condition
- Ran the Damaged condition
- Used the same PZT position
- Used the same frequency
- Used the same excitation type

Then try another damage condition or simulation severity.

ARJUN does not understand my file

Tell ARJUN:

"This data comes from the GARRF Virtual Structural Health Monitoring Laboratory and is simulated educational data."

Then upload the saved experiment file.

PART 24 — SCIENTIFIC LIMITATION

This laboratory is an **educational virtual simulation**.

It does not replace:

- Certified structural inspection
- Real PZT instrumentation
- Laboratory testing
- Engineering judgment
- Field inspection
- Numerical validation
- Structural safety assessment

Real SHM measurements can also be influenced by:

- Temperature
- Humidity
- Sensor bonding
- PZT condition
- Instrumentation
- Cable condition
- Loading
- Boundary conditions
- Environmental noise
- Repeatability

These factors must be considered when working with real structures.

PART 25 — THE GARRF VISION

The goal of this laboratory is not simply to provide an animation.

The goal is to create a pathway:

LEARN

↓

EXPLORE

↓

VIRTUAL EXPERIMENT

↓

OBSERVE

↓

SAVE DATA

↓

TRANSFER DATA

↓

ARJUN ANALYSIS

↓

UNDERSTAND

↓

REAL EXPERIMENT

↓

RESEARCH

This creates a bridge between students who have access to advanced laboratories and students who may not.

FINAL MESSAGE TO THE STUDENT

Do not be afraid to experiment.

Try the controls.

Change the frequency.

Move the PZT.

Try Sine.

Try Sweep.

Try Pulse.

Try Healthy.

Try Crack.

Try different simulation severities.

Try Debonding.

Try different monitoring durations.

Watch the waves.

Watch the waveform.

Watch the FFT.

Making Advanced Structural Health Monitoring Education Accessible to Rural Students

Learn → Explore → Experiment → Observe → Save → Analyze → Understand → Research